MEDGRID Distributed Healthcare Resource Exchange System Architecture Report

System Definition
MEDGRID is a distributed microservice-based healthcare resource exchange and coordination platform that connects multiple healthcare-related facilities—including hospitals, pharmacies, blood banks, and medical equipment/PPE suppliers—through independent service nodes for resource sharing, geo-location-based discovery, and expiry-aware redistribution.

1. System Goals

- Enable healthcare facilities to exchange critical medical resources.
- Support geo-location-based discovery of nearby facilities.
- Monitor inventory expiry and promote redistribution.
- Maintain decentralized facility databases using microservices.
- Coordinate requests between multiple healthcare entities.

2. Participating Facility Types

Hospitals – Request emergency supplies, blood, PPE, and equipment.

Pharmacies – Manage drug inventory and redistribute near-expiry medications.

Blood Banks – Manage blood stock and respond to blood requests.

PPE / Equipment Suppliers – Supply medical equipment and PPE to facilities.

3. High-Level System Architecture

MEDGRID follows a distributed microservice architecture consisting of:

- Frontend Web Application
- API Gateway Layer
- Central Coordination Service
- Independent Facility Microservices
- Separate Local Databases Per Facility

4. Frontend Application

The frontend is responsible for:
- Facility authentication
- Inventory management
- Resource request submission
- Nearby facility discovery
- Expiry monitoring dashboard

• Redistribution workflow

5. API Gateway

The API Gateway acts as the single entry point into the system.

Responsibilities include:
• Routing requests to services
• Authentication validation
• Security middleware
• Request aggregation

6. Coordination Service

The Coordination Service functions as the central network coordinator.

Responsibilities:
• Facility registration and discovery
• Geo-location matching
• Resource request coordination
• Shared inventory metadata management

7. Facility Microservices

Each facility maintains its own independent service and database.

Responsibilities:
• Local inventory management
• Expiry monitoring
• Request handling
• Resource redistribution

8. Geolocation Matching

Facilities are matched using geographic coordinates. The system identifies nearby facilities with available resources and ranks them according to proximity and availability.

9. Expiry Monitoring and Redistribution

MEDGRID monitors inventory expiry dates locally within each facility.

Logic:
If a drug or medical resource is expiring within 60 days, the system flags the item for redistribution.

Nearby facilities can then request the item before expiry.

10. Database Architecture

Central Database Stores:
- Facility profiles
- Coordinates
- Shared inventory summaries
- Resource request metadata

Local Facility Databases Store:
- Actual inventory records
- Expiry dates
- Redistribution transactions
- Internal operational data

11. Technology Stack

- Frontend: React / Next.js
- Backend: Node.js + Express
- Database: PostgreSQL / MySQL
- Authentication: JWT
- ORM: Prisma / Sequelize
- Maps: Leaflet / OpenStreetMap

12. Required System Diagrams

- System Context Diagram
- Use Case Diagram
- High-Level Architecture Diagram
- Deployment Diagram
- Database ER Diagram
- Sequence Diagrams
- Activity Diagrams

13. Project Deliverables

- Full-stack distributed web application
- Multiple facility databases
- Geo-location-based resource discovery
- Expiry monitoring module
- Redistribution coordination workflow
- Academic project documentation

14. Simplified Architecture Representation

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            Frontend Web Application
                       ■
                       ▼
               API Gateway
                       ■
     ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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          ▼                   ▼                   ▼
   Coordination       Hospital Service     Pharmacy Service
      Service
          ■                   ■                   ■
          ▼                   ▼                   ▼
   Blood Bank           PPE Supplier        Local Databases
     Service              Service           (Per Facility)
```

Conclusion

MEDGRID provides a scalable and decentralized healthcare coordination ecosystem that supports resource sharing among hospitals, pharmacies, blood banks, and medical equipment suppliers. The system combines distributed microservice principles, geo-location-based matching, and expiry-aware redistribution into a manageable full-stack academic project architecture.